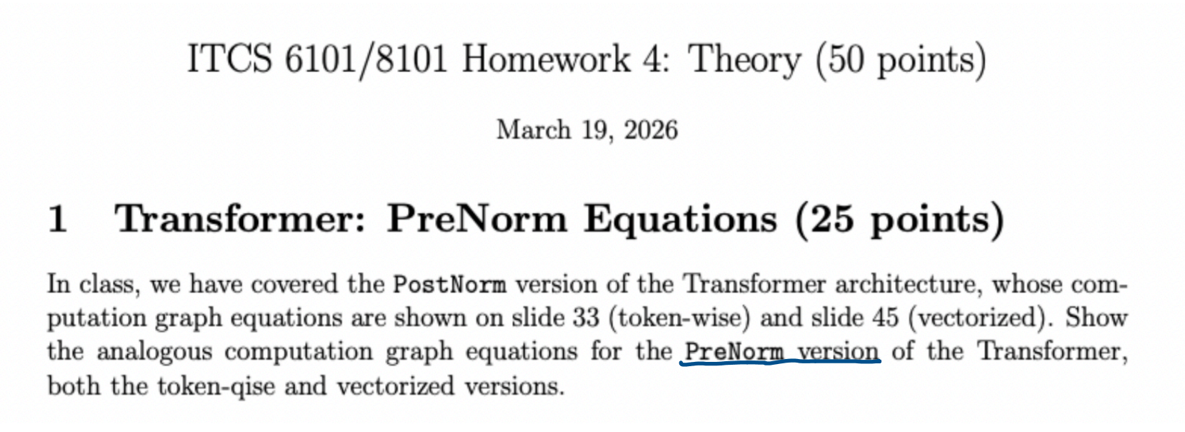
hw05-theory

Monday, March 23, 2026 4:54 PM

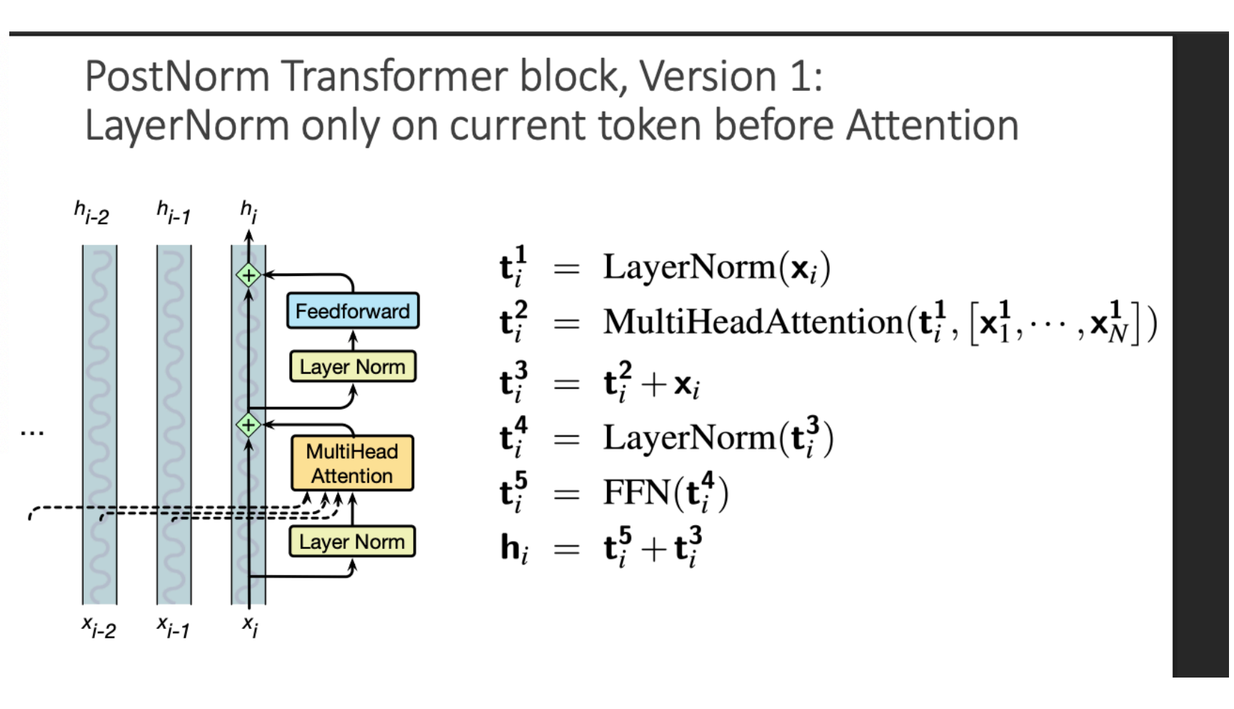


ITCS 6101/8101 Homework 4: Theory (50 points)

March 19, 2026

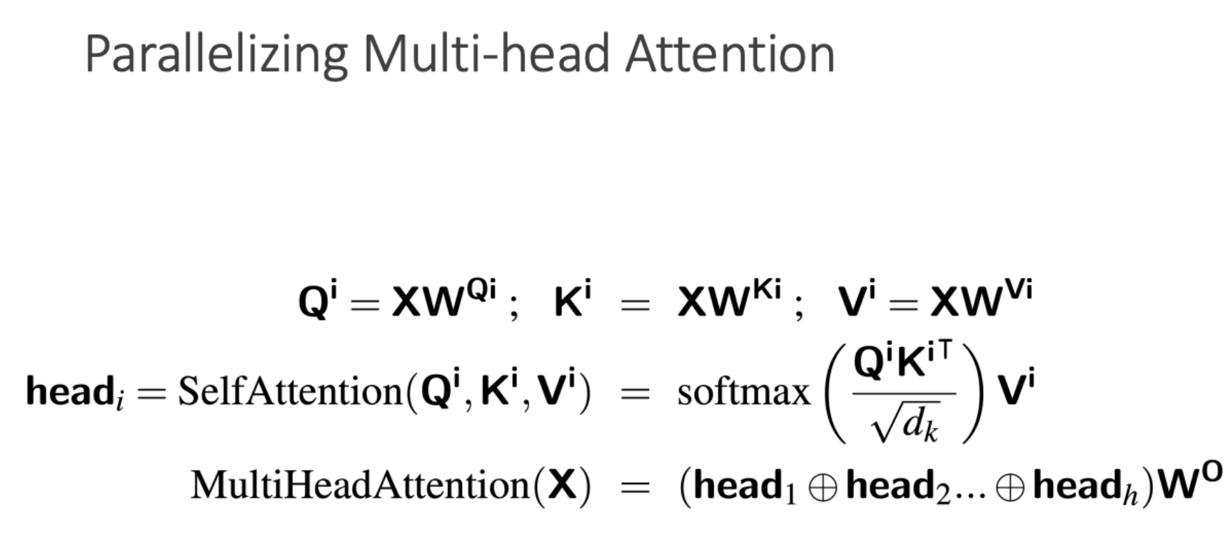
1 Transformer: PreNorm Equations (25 points)

In class, we have covered the `PostNorm` version of the Transformer architecture, whose computation graph equations are shown on slide 33 (token-wise) and slide 45 (vectorized). Show the analogous computation graph equations for the `PreNorm version` of the Transformer, both the token-qise and vectorized versions.



PostNorm Transformer block, Version 1:
LayerNorm only on current token before Attention

$$\begin{aligned}
\mathbf{t}_i^1 &= \text{LayerNorm}(\mathbf{x}_i)\\
\mathbf{t}_i^2 &= \text{MultiHeadAttention}(\mathbf{t}_i^1, [\mathbf{x}_1^1, \cdots, \mathbf{x}_N^1])\\
\mathbf{t}_i^3 &= \mathbf{t}_i^2 + \mathbf{x}_i\\
\mathbf{t}_i^4 &= \text{LayerNorm}(\mathbf{t}_i^3)\\
\mathbf{t}_i^5 &= \text{FFN}(\mathbf{t}_i^4)\\
\mathbf{h}_i &= \mathbf{t}_i^5 + \mathbf{t}_i^3
\end{aligned}$$



Parallelizing Multi-head Attention

$$\begin{aligned}
\mathbf{Q^i} = \mathbf{XW^{Qi}};\quad \mathbf{K^i} &= \mathbf{XW^{Ki}};\quad \mathbf{V^i} = \mathbf{XW^{Vi}}\\
\mathbf{head}_i = \text{SelfAttention}(\mathbf{Q^i}, \mathbf{K^i}, \mathbf{V^i}) &= \text{softmax}\left(\frac{\mathbf{Q^i K^{iT}}}{\sqrt{d_k}}\right)\mathbf{V^i}\\
\text{MultiHeadAttention}(\mathbf{X}) &= (\mathbf{head}_1 \oplus \mathbf{head}_2 \ldots \oplus \mathbf{head}_h)\mathbf{W^O}
\end{aligned}$$

Token-wise :

$t_i^1 = \text{Layer Norm}(x_i)$

$t_i^2 = \text{Multi Head Attention}(t_i^1, [t_1^1, t_2^1 \cdots t_N^1])$

$t_i^3 = t_i^2 + x_i$

$t_i^4 = \text{Layer Norm}(t_i^3)$

$t_i^5 = \text{FFN}(t_i^4)$

$h_i = t_i^5 + t_i^3$

Vectorized :

Let $T^1 = \text{Layer Norm}(X)$

For each head i,

$Q^i = T^1 W^{Q_i} \quad K^i = T^1 W^{K_i} \quad V^i = T^1 W^{V_i}$

$\text{head}_i = \text{Softmax}\left(\frac{Q^i (K^i)^T}{\sqrt{d_k}}\right) V^i$

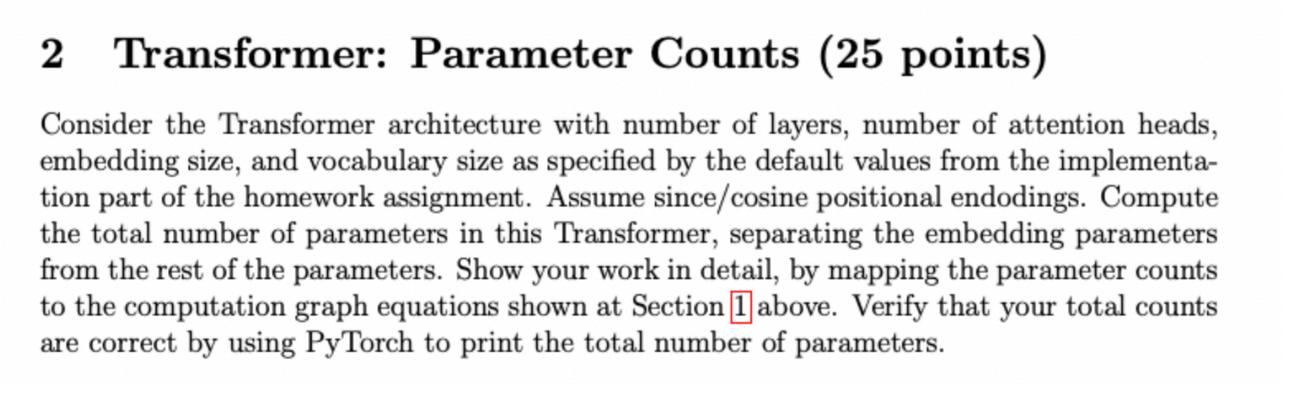
$T^2 = (\text{head}_1 \oplus \text{head}_2 \oplus \cdots \oplus \text{head}_h) W^o$

$T^3 = T^2 + X$

$T^4 = \text{Layer Norm}(T^3)$

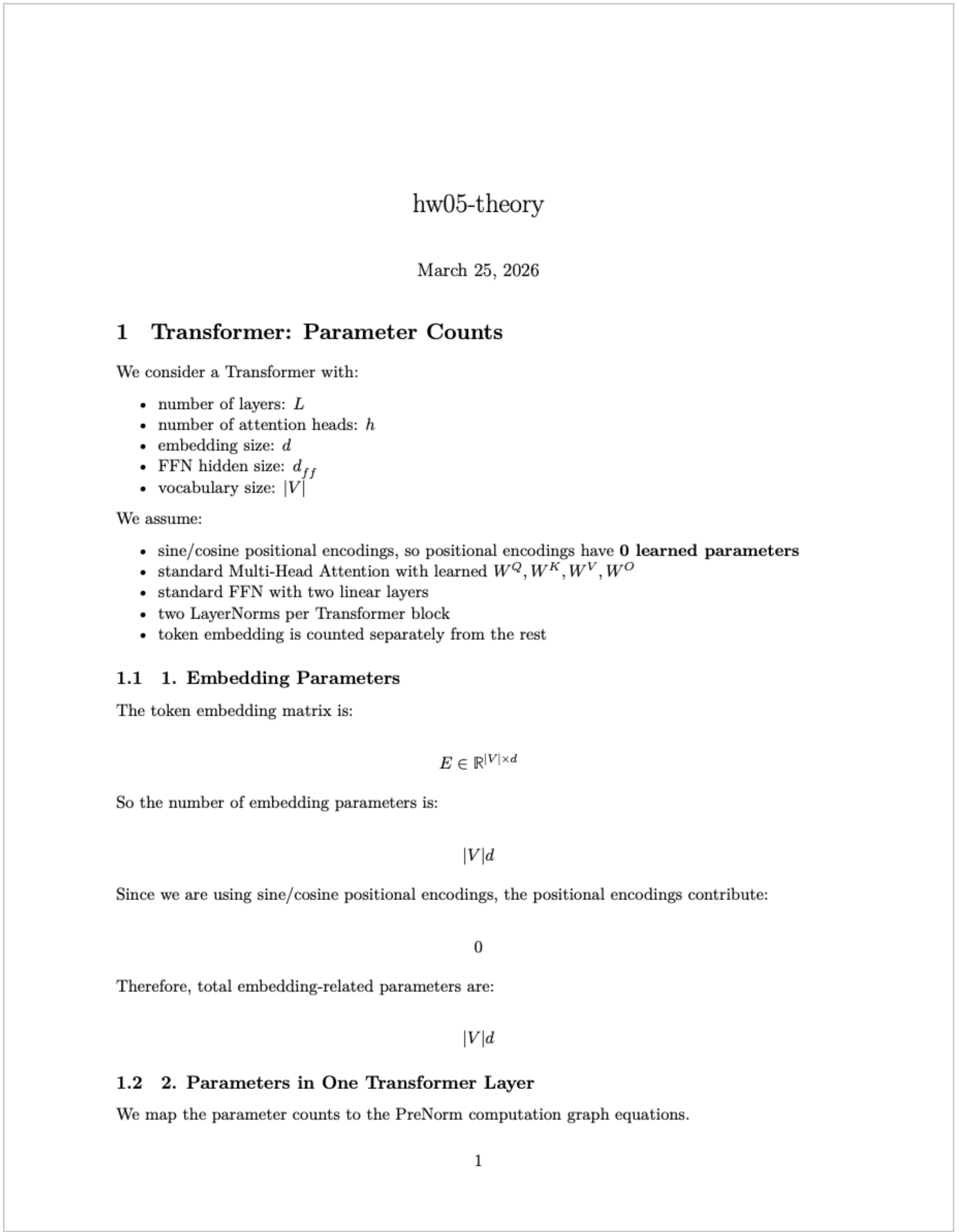
$T^5 = \text{FFN}(T^4)$

$H = T^5 + T^3$



2 Transformer: Parameter Counts (25 points)

Consider the Transformer architecture with number of layers, number of attention heads, embedding size, and vocabulary size as specified by the default values from the implementation part of the homework assignment. Assume since/cosine positional endodings. Compute the total number of parameters in this Transformer, separating the embedding parameters from the rest of the parameters. Show your work in detail, by mapping the parameter counts to the computation graph equations shown at Section 1 above. Verify that your total counts are correct by using PyTorch to print the total number of parameters.



hw05-theory

March 25, 2026

1 Transformer: Parameter Counts

We consider a Transformer with:

- number of layers: L
- number of attention heads: h
- embedding size: d
- FFN hidden size: d_{ff}
- vocabulary size: $|V|$

We assume:

- sine/cosine positional encodings, so positional encodings have **0 learned parameters**
- standard Multi-Head Attention with learned W^Q, W^K, W^V, W^O
- standard FFN with two linear layers
- two LayerNorms per Transformer block
- token embedding is counted separately from the rest

1.1 1. Embedding Parameters

The token embedding matrix is:

$$E \in \mathbb{R}^{|V| \times d}$$

So the number of embedding parameters is:

$$|V|d$$

Since we are using sine/cosine positional encodings, the positional encodings contribute:

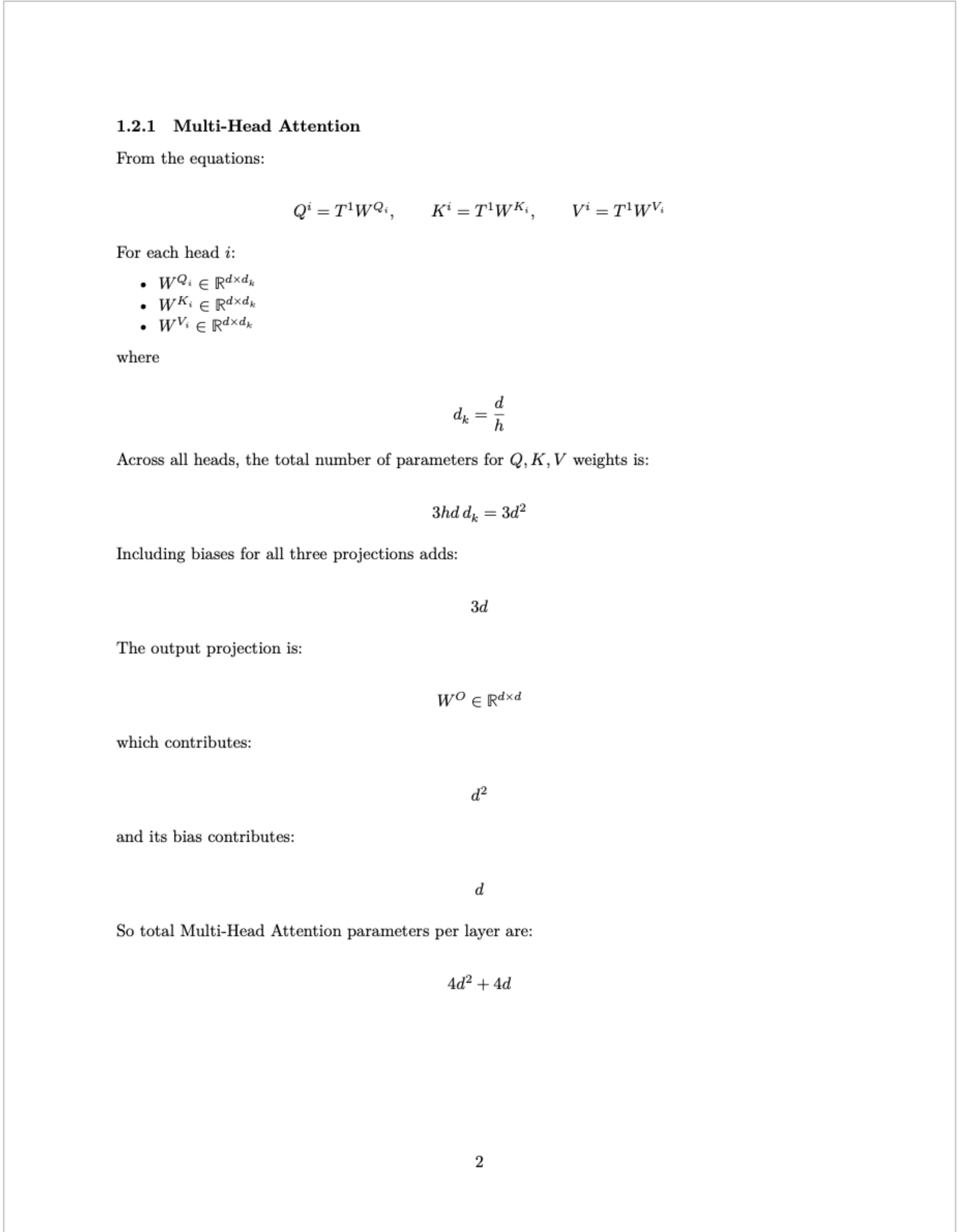
$$0$$

Therefore, total embedding-related parameters are:

$$|V|d$$

1.2 2. Parameters in One Transformer Layer

We map the parameter counts to the PreNorm computation graph equations.



1.2.1 Multi-Head Attention

From the equations:

$$Q^i = T^1 W^{Q_i}, \qquad K^i = T^1 W^{K_i}, \qquad V^i = T^1 W^{V_i}$$

For each head i:

- $W^{Q_i} \in \mathbb{R}^{d \times d_k}$
- $W^{K_i} \in \mathbb{R}^{d \times d_k}$
- $W^{V_i} \in \mathbb{R}^{d \times d_k}$

where

$$d_k = \frac{d}{h}$$

Across all heads, the total number of parameters for Q, K, V weights is:

$$3hd\, d_k = 3d^2$$

Including biases for all three projections adds:

$$3d$$

The output projection is:

$$W^O \in \mathbb{R}^{d \times d}$$

which contributes:

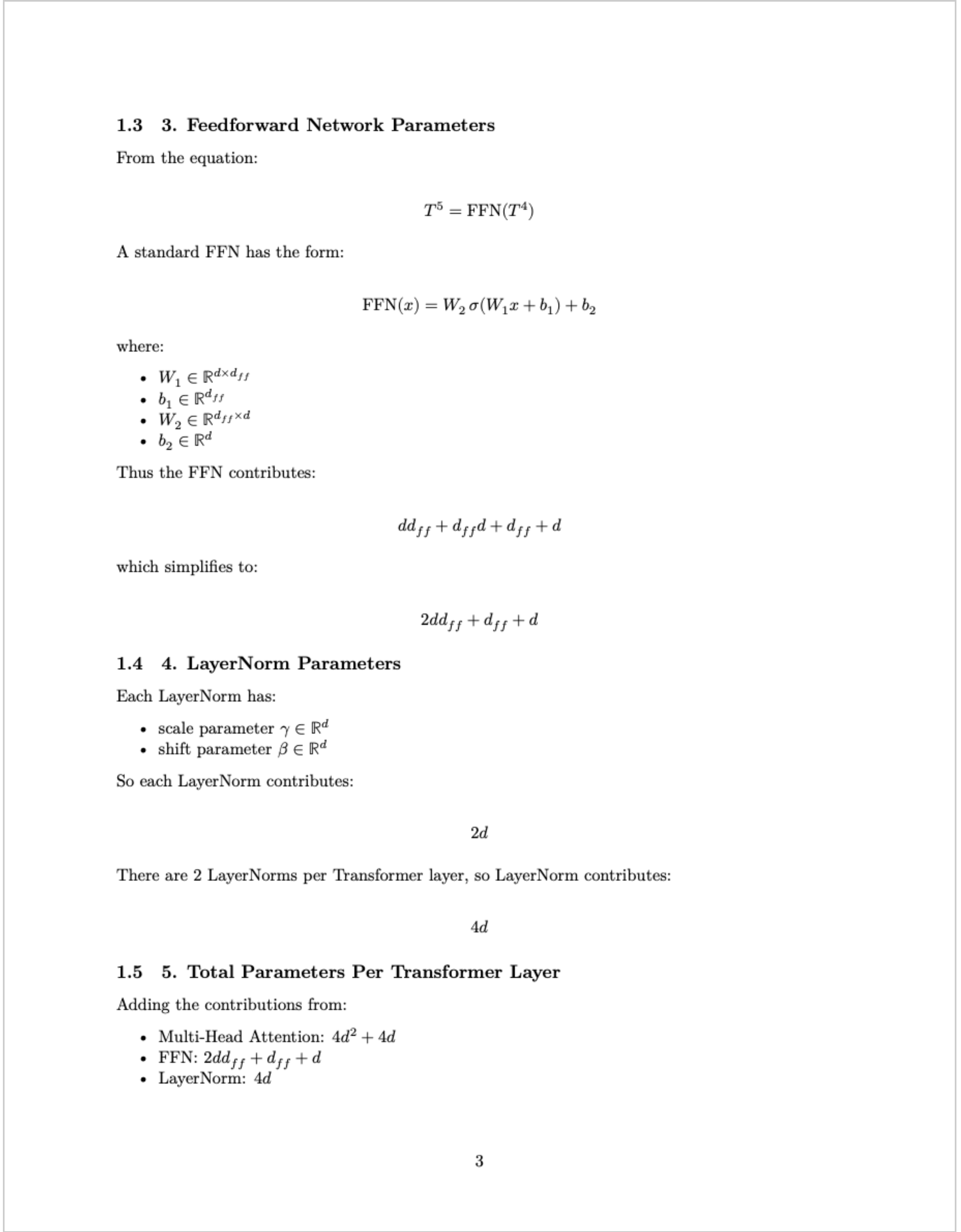
$$d^2$$

and its bias contributes:

$$d$$

So total Multi-Head Attention parameters per layer are:

$$4d^2 + 4d$$



1.3 3. Feedforward Network Parameters

From the equation:

$$T^5 = \text{FFN}(T^4)$$

A standard FFN has the form:

$$\text{FFN}(x) = W_2\, \sigma(W_1 x + b_1) + b_2$$

where:

- $W_1 \in \mathbb{R}^{d \times d_{ff}}$
- $b_1 \in \mathbb{R}^{d_{ff}}$
- $W_2 \in \mathbb{R}^{d_{ff} \times d}$
- $b_2 \in \mathbb{R}^d$

Thus the FFN contributes:

$$dd_{ff} + d_{ff}d + d_{ff} + d$$

which simplifies to:

$$2dd_{ff} + d_{ff} + d$$

1.4 4. LayerNorm Parameters

Each LayerNorm has:

- scale parameter $\gamma \in \mathbb{R}^d$
- shift parameter $\beta \in \mathbb{R}^d$

So each LayerNorm contributes:

$$2d$$

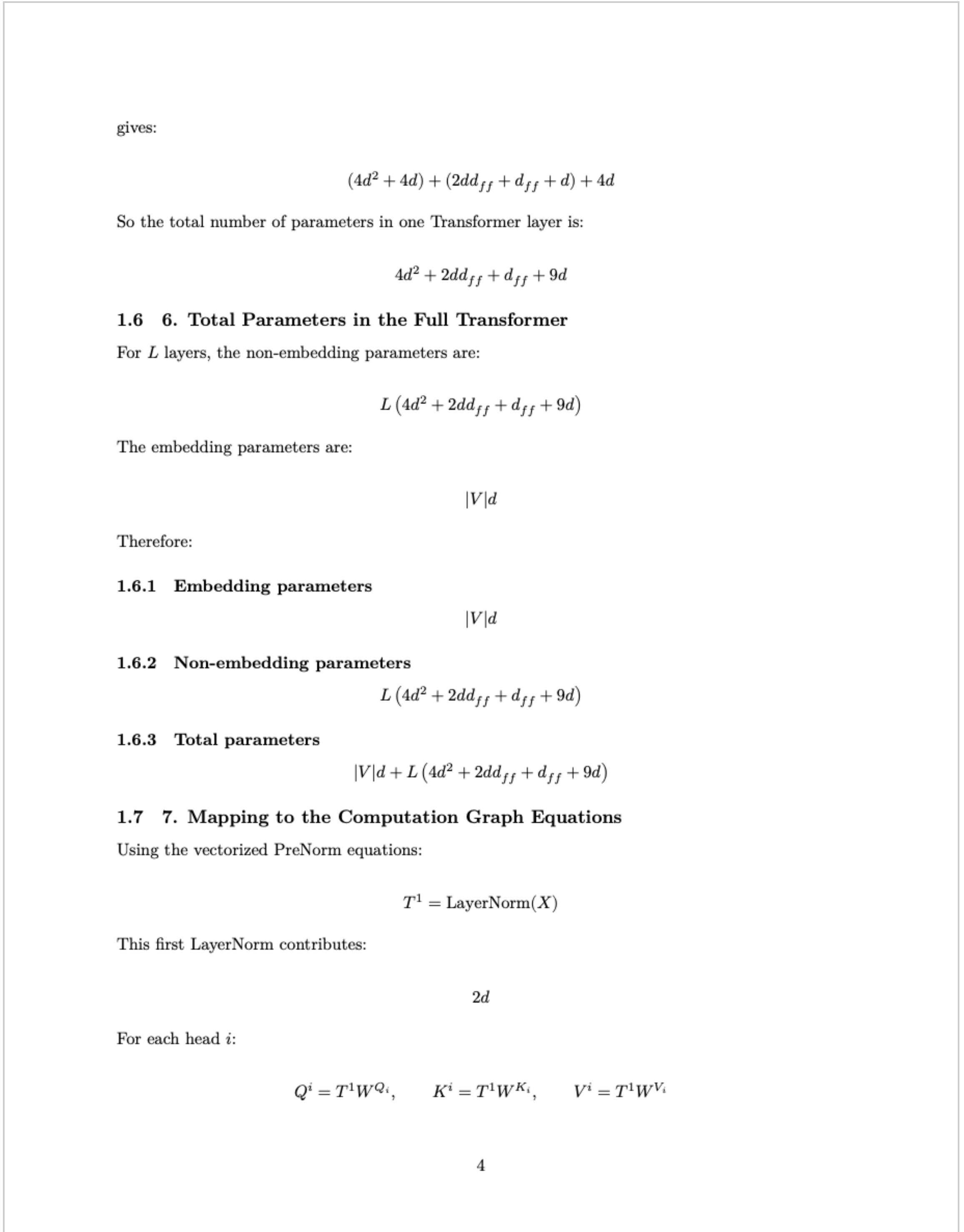
There are 2 LayerNorms per Transformer layer, so LayerNorm contributes:

$$4d$$

1.5 5. Total Parameters Per Transformer Layer

Adding the contributions from:

- Multi-Head Attention: $4d^2 + 4d$
- FFN: $2dd_{ff} + d_{ff} + d$
- LayerNorm: $4d$



gives:

$$(4d^2 + 4d) + (2dd_{ff} + d_{ff} + d) + 4d$$

So the total number of parameters in one Transformer layer is:

$$4d^2 + 2dd_{ff} + d_{ff} + 9d$$

1.6 6. Total Parameters in the Full Transformer

For L layers, the non-embedding parameters are:

$$L\left(4d^2 + 2dd_{ff} + d_{ff} + 9d\right)$$

The embedding parameters are:

$$|V|d$$

Therefore:

1.6.1 Embedding parameters

$$|V|d$$

1.6.2 Non-embedding parameters

$$L\left(4d^2 + 2dd_{ff} + d_{ff} + 9d\right)$$

1.6.3 Total parameters

$$|V|d + L\left(4d^2 + 2dd_{ff} + d_{ff} + 9d\right)$$

1.7 7. Mapping to the Computation Graph Equations

Using the vectorized PreNorm equations:

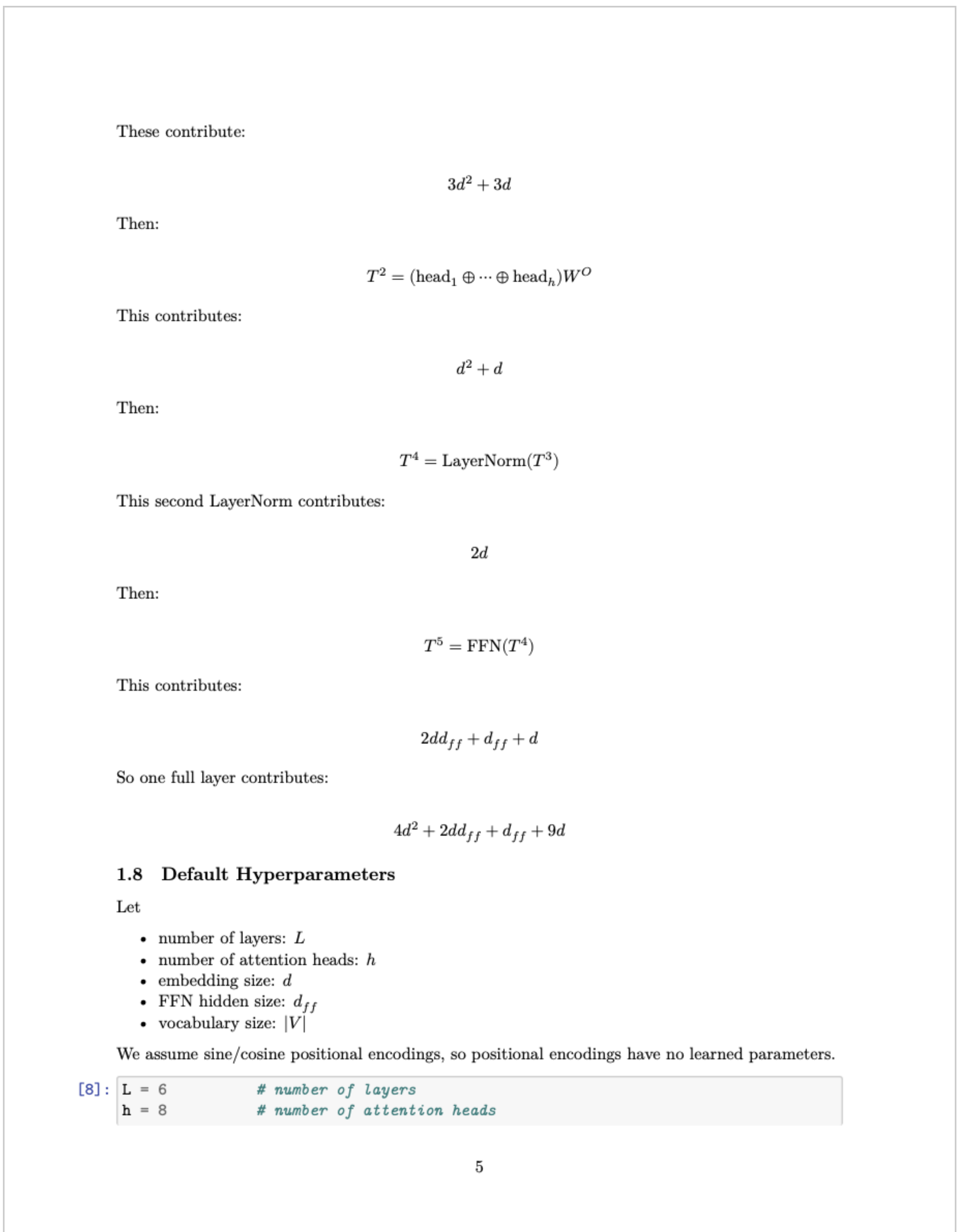
$$T^1 = \text{LayerNorm}(X)$$

This first LayerNorm contributes:

$$2d$$

For each head i:

$$Q^i = T^1 W^{Q_i}, \qquad K^i = T^1 W^{K_i}, \qquad V^i = T^1 W^{V_i}$$



These contribute:

$$3d^2 + 3d$$

Then:

$$T^2 = (\text{head}_1 \oplus \cdots \oplus \text{head}_h) W^O$$

This contributes:

$$d^2 + d$$

Then:

$$T^4 = \text{LayerNorm}(T^3)$$

This second LayerNorm contributes:

$$2d$$

Then:

$$T^5 = \text{FFN}(T^4)$$

This contributes:

$$2dd_{ff} + d_{ff} + d$$

So one full layer contributes:

$$4d^2 + 2dd_{ff} + d_{ff} + 9d$$

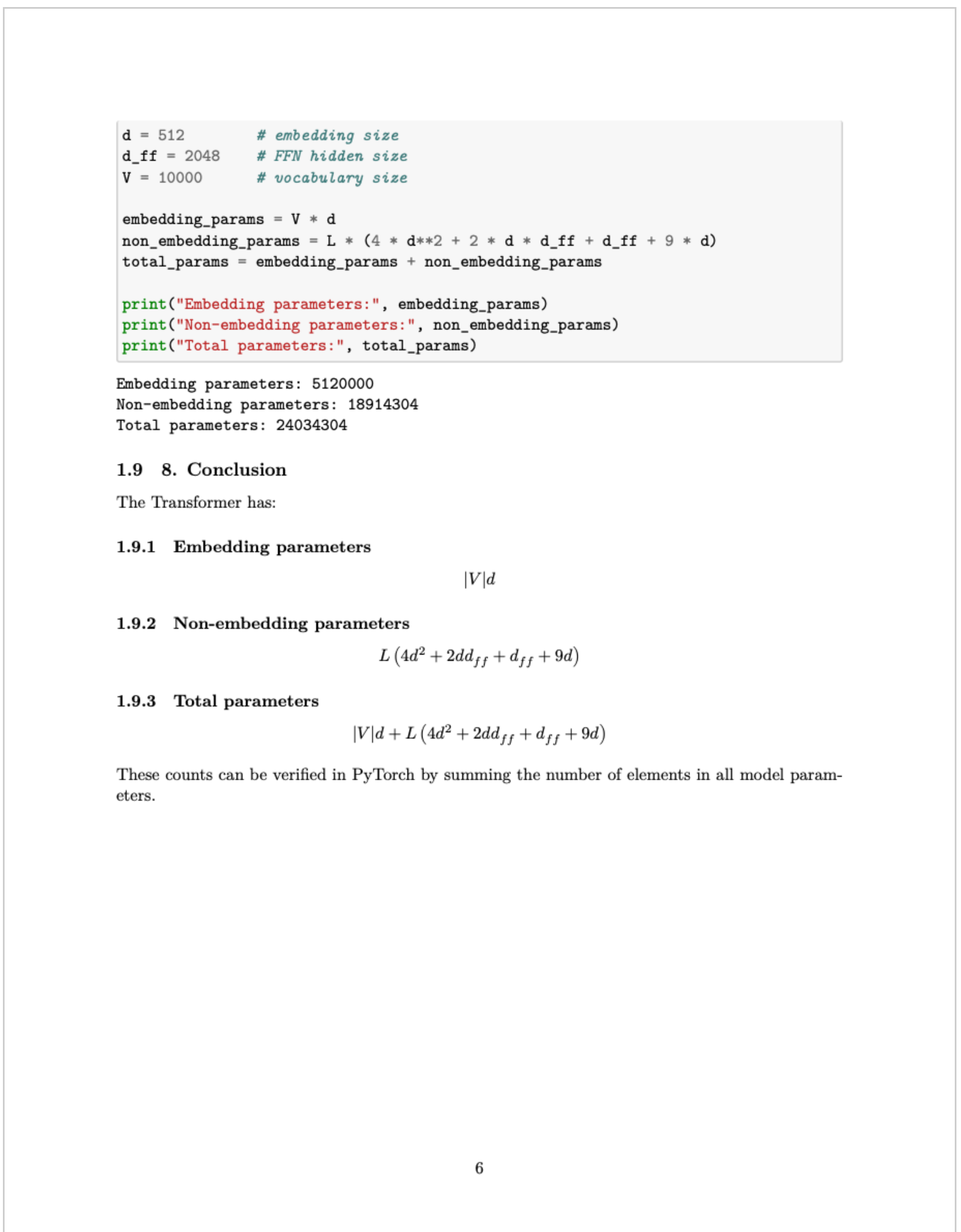
1.8 Default Hyperparameters

Let

- number of layers: L
- number of attention heads: h
- embedding size: d
- FFN hidden size: d_{ff}
- vocabulary size: $|V|$

We assume sine/cosine positional encodings, so positional encodings have no learned parameters.

```
[8]: L = 6          # number of layers
     h = 8          # number of attention heads
```



```
d = 512        # embedding size
d_ff = 2048    # FFN hidden size
V = 10000      # vocabulary size

embedding_params = V * d
non_embedding_params = L * (4 * d**2 + 2 * d * d_ff + d_ff + 9 * d)
total_params = embedding_params + non_embedding_params

print("Embedding parameters:", embedding_params)
print("Non-embedding parameters:", non_embedding_params)
print("Total parameters:", total_params)
```

```
Embedding parameters: 5120000
Non-embedding parameters: 18914304
Total parameters: 24034304
```

1.9 8. Conclusion

The Transformer has:

1.9.1 Embedding parameters

$$|V|d$$

1.9.2 Non-embedding parameters

$$L\left(4d^2 + 2dd_{ff} + d_{ff} + 9d\right)$$

1.9.3 Total parameters

$$|V|d + L\left(4d^2 + 2dd_{ff} + d_{ff} + 9d\right)$$

These counts can be verified in PyTorch by summing the number of elements in all model parameters.